

Green Campus Sustainability Intelligence Platform

Easy explanation guide for demo presentation and stakeholder walkthrough

Parameter	Value
Emission factor	0.82 kg CO2 per kWh
Energy cost	8.0 per kWh
Energy benchmark	100000
Water benchmark	20000
Waste benchmark	30000
Green Index weights	Energy 0.5, Water 0.3, Waste 0.2
Alert thresholds	15% standard, 30% high

1. What this project does

- Green Campus is a sustainability dashboard for educational campuses.
- It combines electricity, water, waste, and solar information in one place.
- The platform helps a user see the current situation, compare buildings, understand carbon impact, and plan future usage.

2. How to explain the website to someone else

- Start from the top hero section and explain that this is the command center.
- Then show the current scope, forecast mode, and data controls.
- Next explain that the first cards are quick KPI summaries, the next sections explain carbon and solar, and the lower sections provide AI-style planning support.

3. Website demo flow step by step

- Step 1: Login to the application.
- Step 2: Explain that the dashboard is filter-based and can be viewed for the whole campus or a single building.
- Step 3: Show KPI cards for energy, water, waste, solar, net energy, and Green Index.
- Step 4: Open carbon and solar sections and explain how solar reduces grid demand.
- Step 5: Show alerts, trends, and the resource mix chart.
- Step 6: Show AI insights, risk, forecast, and scenario simulation.
- Step 7: Show assumptions and terminology so non-technical users understand the values.
- Step 8: Show import, export, and reset workflow.

4. Exact calculation logic used in the system

- Gross Energy = total of all energy_kwh records in the selected scope.
- Solar = total of all solar_kwh records in the selected scope.
- Net Energy = $\max(\text{Gross Energy} - \text{Solar}, 0)$.
- Export to Grid = $\max(\text{Solar} - \text{Gross Energy}, 0)$.
- Carbon = Net Energy x 0.82 kg CO₂ per kWh.
- Gross Carbon = Gross Energy x 0.82.
- Solar Avoided Carbon = $\min(\text{Solar}, \text{Gross Energy}) \times 0.82$.

4A. Simple meaning of important terms

- KPI means Key Performance Indicator. It is a quick important number shown on the dashboard card so a user can understand the situation fast.
- Gross Energy means the total electricity used before subtracting solar power.
- Net Energy means the electricity still needed after using solar power.
- Solar Offset means how much electricity demand was covered by solar instead of the grid.
- Gross Carbon means carbon linked to full electricity demand before solar benefit is considered.
- Net Carbon means carbon linked only to the remaining grid electricity after solar support.
- Solar Avoided Carbon means the carbon saved because solar reduced the amount of grid electricity needed.
- Benchmark means a reference target or comparison value. It helps the system decide whether current usage is low, acceptable, or high.
- Efficiency Score or Weighted Efficiency means the combined pressure from energy, water, and waste after giving each one a fixed importance.
- Green Index means the final sustainability score shown in an easy percentage form. Higher is better.
- Forecast Granularity means the time level used in prediction, such as daily, monthly, yearly, or seasonal.
- Scenario Simulator means a what-if tool that shows what may happen if energy, water, waste, or solar values change.
- Alert Threshold means the change percentage after which the system starts calling something unusual.
- Export Potential means solar generation that may remain after meeting campus demand and could be sent to storage or the grid.

5. Green Index calculation step by step

- Energy benchmark = 100000
- Water benchmark = 20000
- Waste benchmark = 30000
- Energy Score = $\min(\text{Net Energy} / 100000, 1)$
- Water Score = $\min(\text{Water} / 20000, 1)$
- Waste Score = $\min(\text{Waste} / 30000, 1)$

- $\text{Weighted Efficiency} = 0.5 \times \text{Energy Score} + 0.3 \times \text{Water Score} + 0.2 \times \text{Waste Score}$
- $\text{Green Index} = (1 - \text{Weighted Efficiency}) \times 100$

6. Forecast logic in easy words

- The system supports daily, monthly, yearly, and seasonal views.
- It groups historical records according to the selected mode.
- Then it looks for recurring patterns in that grouping and uses average cycle behavior to generate the next periods.
- This is practical and easy to explain during a demo because it uses historical campus behavior directly.

7. Seasonal logic and college-specific example

- The platform knows that some months have lower occupancy.
- For example, June has lower academic activity, so demand is expected to reduce.
- That means energy and water can fall, while solar can cover a larger percentage of demand.
- If solar becomes greater than demand, the dashboard shows export potential.

8. Import, export, and reset

- Import Excel replaces the current workbook data with a fresh dataset.
- Export Excel downloads the live system data with summary and raw sheets.
- Reset Data clears resource records when a fresh demo or new dataset is needed.

9. Current strengths of the project

- Strong end-to-end flow from login to dashboard to data actions.
- Good campus-specific logic for occupancy, solar offset, and utility planning.
- Useful mix of operational metrics and AI-style recommendation features.

10. Current improvement opportunities

- The frontend can still be improved further by moving more repeated card styles into reusable UI building blocks.
- The backend can be split more cleanly into service-layer calculations and thinner routers.
- Future versions can add PDF management reports, smarter anomaly detection, and occupancy-linked forecasting.